SKILL BUILDERS · UNIT 8 · SCORE 3

# Transformations: Practice Set

NC.8.G.2–4

**ANSWER KEY**

Do one problem at a time. Show your steps. Circle the letter or write the number.

**1** Score 3 NC.8.G.3

Translate left 2 and down 3: the point **(2, 5)**. Where does it land?

- (A) (−1, 3)
- **(B) (0, 2)**
- (C) (4, 8)
- (D) (0, 8)

**2** Score 3 NC.8.G.3

Reflect over the y-axis: the point **(−3, 4)**. Where does it land?

- (A) (4, −3)
- (B) (−3, −4)
- (C) (3, −4)
- **(D) (3, 4)**

**3** Score 3 NC.8.G.3

Reflect over the x-axis: the point **(6, −1)**. Where does it land?

- **(A) (6, 1)**
- (B) (−6, 1)
- (C) (−1, 6)
- (D) (−6, −1)

**4** Score 3 NC.8.G.3

Rotate 180° about the origin: the point **(2, 7)**. Where does it land?

- **(A) (−2, −7)**
- (B) (−2, 7)
- (C) (−7, 2)
- (D) (7, 2)

**5** Score 3 NC.8.G.4

Dilate by k = 4: the point **(1, 3)**. Where does it land?

- (A) (4, 3)
- **(B) (4, 12)**
- (C) (5, 7)
- (D) (0.25, 0.75)

**6** Score 3 NC.8.G.3

Rotate 90° counterclockwise about the origin: the point **(5, 5)**. Where does it land?

- (A) (−5, −5)
- (B) (5, −5)
- (C) (5, 5)
- **(D) (−5, 5)**

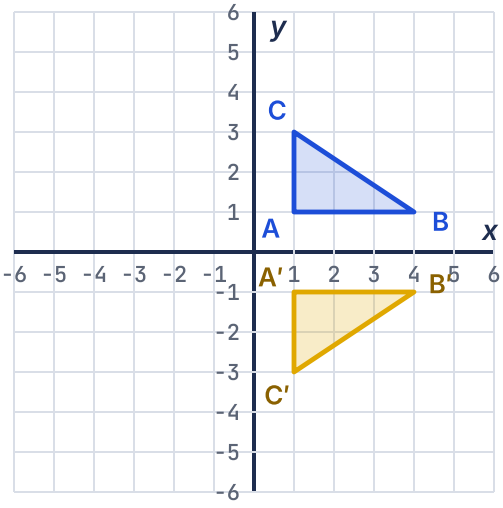
**7** Score 3 NC.8.G.4

Dilate by k = 1/2: the point **(4, −6)**. Where does it land?

- Ⓐ (4.5, −5.5)
- Ⓑ **(2, −3)**
- Ⓒ (2, 3)
- Ⓓ (8, −12)

**8** Score 3 NC.8.G.3

Which move takes the blue triangle to the gold triangle?




- Ⓐ Translation down 4
- Ⓑ **Reflection over the x-axis**
- Ⓒ Dilation by k = 2
- Ⓓ Rotation 180°

**9** Score 3 NC.8.G.2

After a translation, a reflection and a rotation, the image is...

- Ⓐ **congruent: same size and shape**
- Ⓑ similar but smaller
- Ⓒ similar but bigger

**10** Score 3 NC.8.G.3

First translate right 3, then reflect over the x-axis: the point **(1, 2)**. Where does it land?

- Ⓐ (4, 2)
- Ⓑ **(4, −2)**
- Ⓒ (−4, −2)
- Ⓓ (1, −5)

1 `(1, 2) → right 3 → (4, 2)`

2 `(4, 2) → over x-axis → (4, -2)`

**11** Score 3 NC.8.G.3

**Find the mistake.** Mia reflected (2, 6) over the x-axis and got `(-2, 6)`. What went wrong?

**A Mia changed x. Over the x-axis, y changes sign: (2, −6).**

B Mia is right.

C The numbers should switch: (6, 2).

D Both signs should change: (−2, −6).